

TL;DR – Too long, dint read

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Explain LLM algorithms

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LLM Algorithms — Simple Beginner Explanation

LLM (Large Language Model) algorithms are the techniques that help AI understand language, learn patterns, and generate responses like humans.

Think of an LLM like a super smart prediction engine.

It reads huge amounts of text and learns:

- how words connect
- sentence patterns
- meaning
- context
- reasoning styles

Main Algorithms Used in LLMs

1. Transformer Algorithm (Most Important)

Transformer

This is the core alaoirgithm behind modern LLMs like:

+ Ask anything

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using something called **Attention Mechanism**.

2. Attention Mechanism

Attention Mechanism

This helps the model focus on important words.

Example:

Sentence:

"Ravi went to the bank to deposit money."

Attention helps model understand:

- "bank" = financial bank
- not river bank

because of words like:

- deposit
- money

3. Neural Networks

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TL;DR

TL;DR — LLM Algorithms

- **Tokenization** → Breaks sentence into small pieces (tokens)
- **Embeddings** → Converts words into numerical vectors
- **Transformer** → Main architecture that processes language
- **Attention / Self-Attention** → Finds which words are important to each other
- **Neural Networks** → Learns patterns from huge text data
- **Backpropagation** → Learns from mistakes
- **Gradient Descent** → Reduces prediction errors step by step

Simple Flow

Text → Tokens → Vectors → Attention → Prediction → Response

Main goal of all these algorithms:

Help AI understand context and predict the next best word intelligently.

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TL;DR gives gist of previous prompt output which was too informative. But never misses a point.

ELI – Explain Like I am

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ELI15

ELI15 — LLM Algorithms

Imagine an AI like a super smart student who has read almost the entire internet 🤖

But...

it does NOT "understand" like humans.

It works mainly by:

finding patterns and predicting the next best word.

Example

You type:

"I am feeling very..."

AI thinks:

- happy?
- tired?
- excited?

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Main Algorithms in Simple Terms

1. Tokenization

Sentence ni small pieces ga cut chestundi.

Example:

"I love AI"

becomes:

- I
- love
- AI

These pieces = tokens.

2. Embeddings

Words ni numbers ga convert chestundi.

Because computers understand numbers, not English.

Example:

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This happens billions of times.

Final Flow

- User text
- broken into tokens
- converted into vectors
- Transformer checks relationships
- Attention finds important words
- AI predicts next word
- response generated

Simple One-Line Understanding

LLM algorithms are systems that help AI:

convert words into numbers, understand relationships between words, and predict the next best response intelligently.

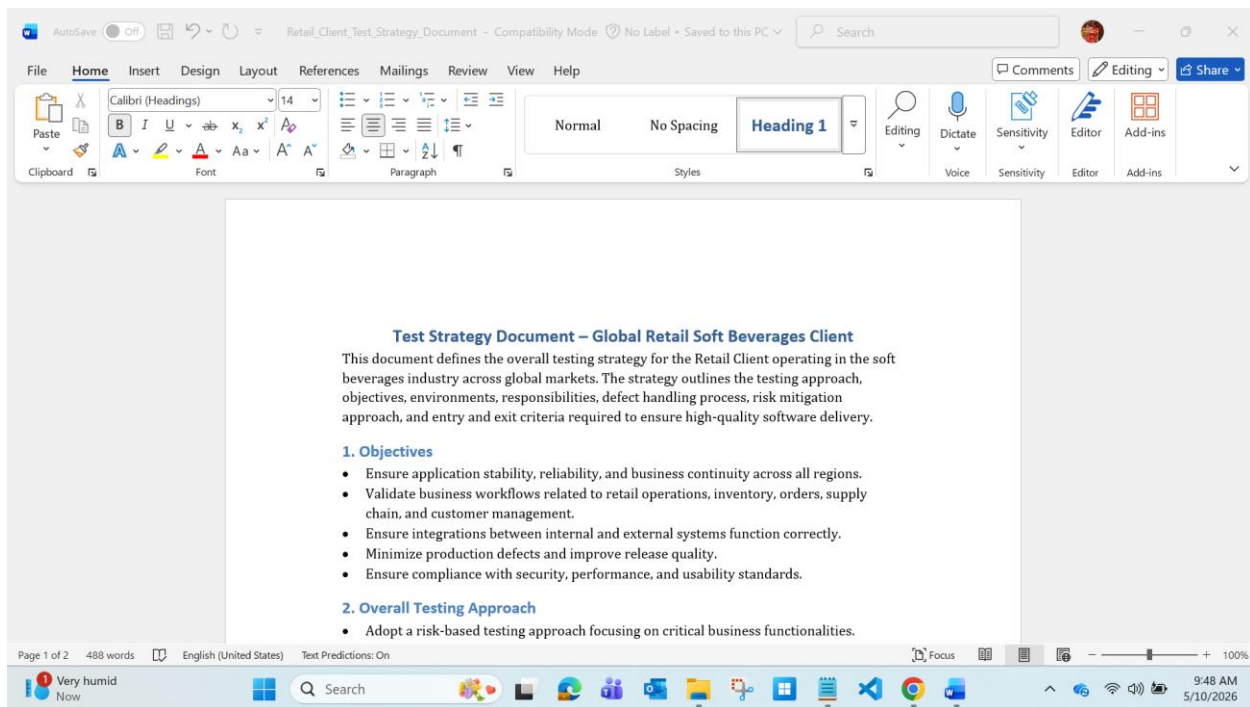
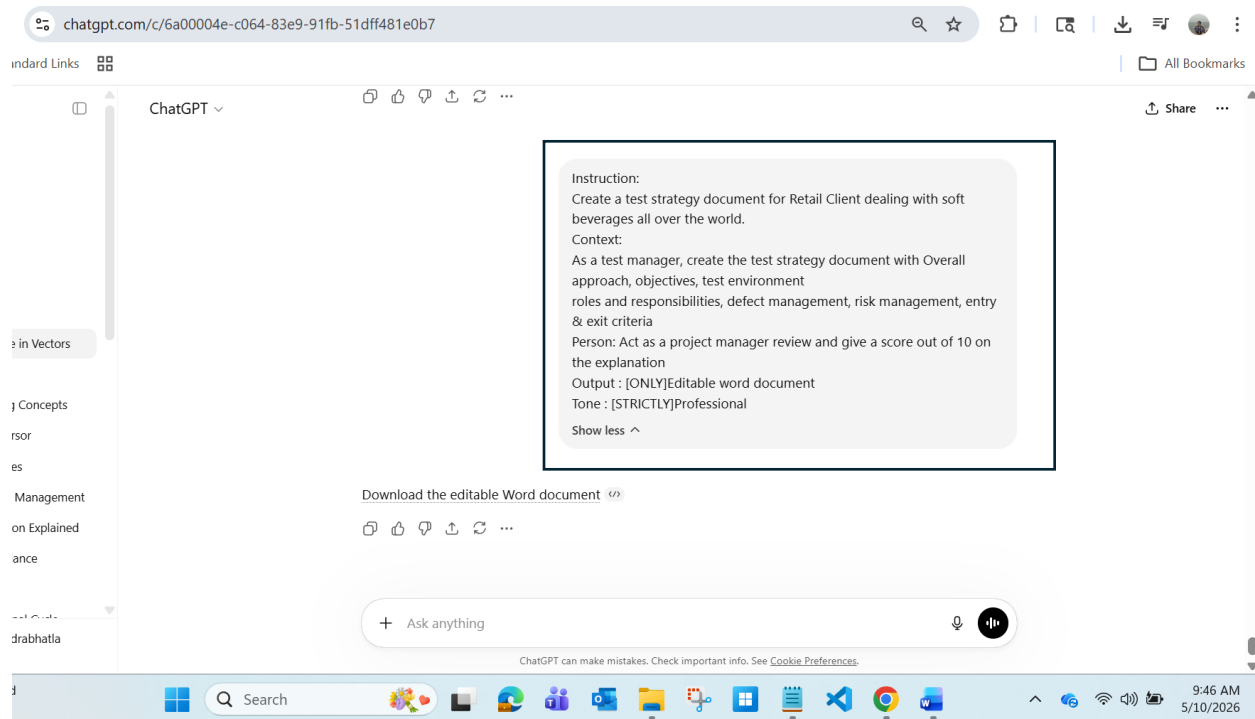
+ Ask anything

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Prompt ELI explains the output in meaningful bits and pieces without missing the flow. This explanation is not too heavy as seen in the generic prompt nor it is brief like in TL;DR. ELI gives understandable short examples for each concept. All one liner.

Structured Reasoning Prompt: All details/instructions given in the prompt at a once.



Given the link of this doc in Github. Structured Reasoning Prompt almost follows ICEPOT framework . All the details are constraints are given in the prompt.

Step by Step Prompting: It is all about “how AI should think”

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Create the test strategy document in Professional Tone
Step1: Get the domain details
Step2: Get the number of testers
Step3: What is Test objective
Step4: Entry Criteria
Step5: Exit Criteria
Step6: How defects are managed?
Step7: Roles and responsibilities
Step8: Output expected in?

AI Response (Step-by-Step Requirement Gathering)

Step 1: Get the Domain Details

Please provide:

- Project domain (Retail, Banking, Healthcare, Insurance, etc.)
- Application type (Web, Mobile, API, Desktop)
- Business overview

Example:



Global Retail Client dealing with soft beverages

+ Ask anything



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Step 2: Get the Number of Testers

Please provide:

- Total QA team size
- Manual testers count
- Automation testers count
- QA Leads/Test Managers involved

Example:

10 Manual Testers, 4 Automation Testers, 1 QA Lead

Step 3: What is the Test Objective?

Please provide:

- Main business goals of testing
- Types of testing required

Example:

- Functional Testing
- Regression Testing
- Integration Testing
- Performance Testing
- UAT Support

...

Step 4: Entry Criteria

Please provide:

- Conditions required before testing starts

Example:

- Requirements approved
 - Test environment ready
 - Build deployed successfully
 - Test data available
-

Step 5: Exit Criteria

Please provide:

- Conditions required before testing closure

Example:

- All critical defects closed
- Test cases executed
- Regression completed
- UAT signoff received



Step 6: How are Defects Managed?

Please provide:

- Defect tracking tool
- Severity/Priority process
- Defect lifecycle details

Example:

- Jira used for defect tracking
 - Daily defect triage meetings conducted
-

Step 7: Roles and Responsibilities

Please provide:

- Team structure and responsibilities

Example:

- Test Manager
- QA Lead
- QA Analysts
- Automation Engineers
- Developers
- Business Stakeholders



Step 8: Output Expected In?

Please confirm output format:

Options:

- Editable Word document
- PDF
- Excel
- PPT
- Markdown document

Once all details are provided, the professional Test Strategy document can be generated.



Retail
27 , Manual - 20 , Automation -7, QA Leads -2 , Manager -1
Functional, Integration & UAT Support
Requirements Approved & Test Env Ready
All P1 & P2 defects closed, Test Cases Executed
Defect Tracking tool Devops

- PPT
- Markdown document

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PDF

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PDF

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Do not include the project manager review in the downloadable PDF

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Step by Step prompting help in giving a clear instruction to the LLM on what has to be done when the output has many sections to be filled. Also these instructions can be given in one go or really one step after other which makes it a chain of prompts.